

Multi-Asset Portfolio Optimization Pipeline

Data collection, preprocessing, asset selection, SI weight optimization, and walk-forward testing.

Base currency: THB

Ledoit-Wolf Covariance

HRP + ACO Selection

PSO / ACO+EBGWO Weights

Data Collection

Executed Universe

- 30 assets in `config/assets.csv`.
- Thai equities, US-listed equities, and bonds.
- SPY appended automatically as S&P 500 benchmark.
- USD assets converted to THB.

Requested Expansion

- S&P 500 assets.
- Thai assets.
- Chinese assets.
- ETC and Crypto.

Chinese, ETC, and Crypto are not in the current executed `assets.csv`.

Train / Test Design

Requested Windows

Window	Start	End
Train	2015-01-01	2024-01-01
Test	2025-01-01	2026-01-01

Current Pipeline Config

Setting	Value
Start date	2015-01-01
In-sample end	2024-01-01
Out-of-sample end	2026-03-20
Transaction cost	0.1% turnover

Implementation note: `backtest.py` defaults to rolling 3-year lookback and quarterly step, not 4-year / 1-year.

Data Preprocessing

365d

MISSING GAP CUTOFF

Drop assets with consecutive missing values longer than 365 days.

log

RETURN TRANSFORM

Convert prices to log returns: $\log(P_t / P_{t-1})$.

LW

COVARIANCE

Use Ledoit-Wolf shrinkage covariance with PyTorch device auto-detection.

Currency Handling

USD-denominated assets are converted into THB using $\text{USDTHB}=X$; if FX download fails, the fallback rate is 35.0 THB/USD.

Asset Selection Strategies

Strategy	Idea
Sharpe Ratio	Select strongest individual risk-adjusted performers.
Diversification Score	Reward high Sharpe and lower average correlation.
Ledoit-Wolf + Diversification	Use robust covariance and correlation estimates.
HRP + Max Sharpe per Cluster	Cluster similar assets, then select best Sharpe representative.
HRP + Max Diversification per Cluster	Cluster first, then choose best diversification representative.
2D-ACO / 2D-ACO + Cluster	Discrete pheromone-guided asset selection, optionally constrained by clusters.

Weight Optimization and Objective

Representation

- ACO selection: integer / discrete path encoding.
- PSO, EBGWO, SI/DE-style methods: float weight vectors.
- Current optimization is long-only.

Objective

Maximize Sharpe Ratio with entropy reward and concentration / constraint penalties.

Fitness uses Ledoit-Wolf volatility estimates and annualized return assumptions.

PSO

Particles search weight space using personal and global best portfolios.

ACO

Pheromone trails and Sharpe heuristics bias asset paths.

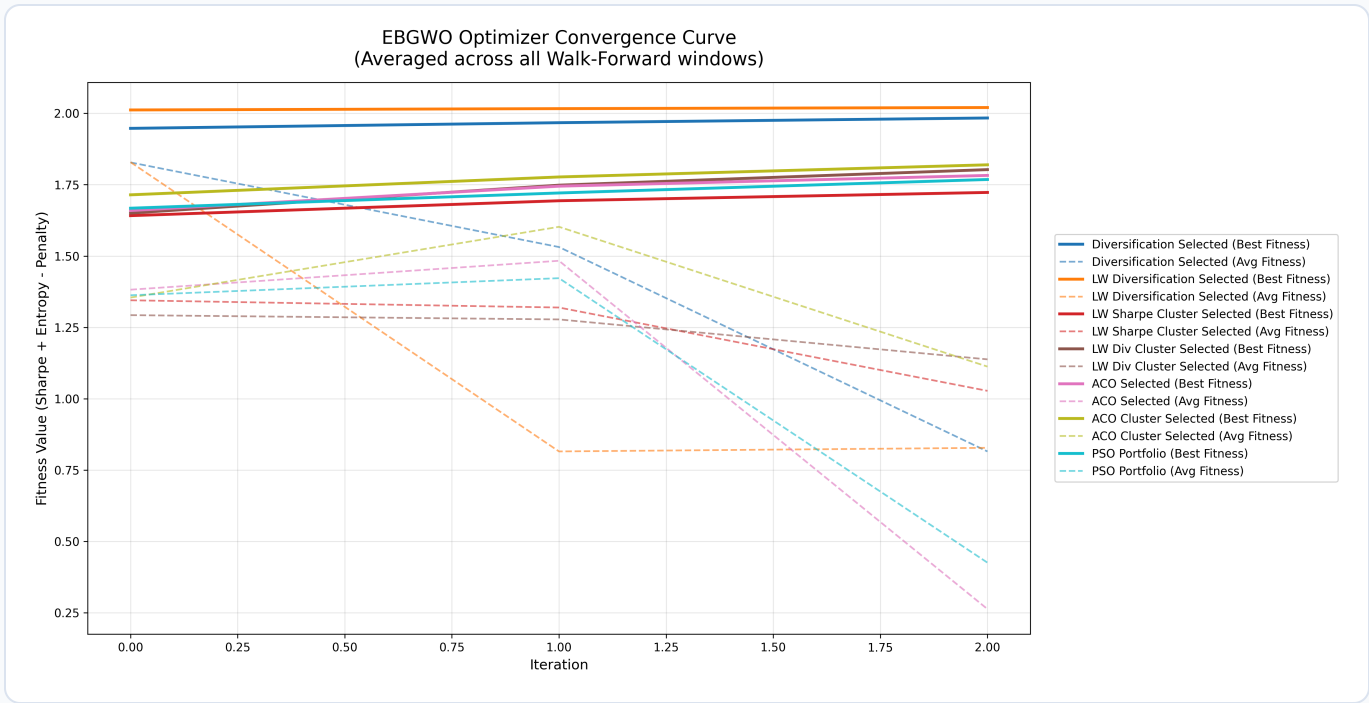
EBGWO

Wolf population follows leading solutions while preserving exploration.

Constraints and Bonus Features

Item	Requested	Current status
Minimum assets	5	Implemented as lower bound through <code>`max(5, len(valid_stocks)//2)`</code> .
Maximum assets	15	Needs explicit enforcement if strict max 15 is required.
Sector constraint	Bonus	Available in dynamic ACO+EBGWO via sector labels.
Short sell	Bonus	Not enabled in current long-only encoding.
Transaction cost	Required realism	0.1% turnover cost applied at each rebalance.

Convergence and Stability



How Stability Is Reported

- Latest PSO run: 2 trials per strategy.
- Latest ACO+EBGWO run: 2 trials per strategy.
- Tables report mean +- standard deviation.
- Convergence tracks best and average fitness.

Evaluation Metrics

Sharpe

REPORTED

Risk-adjusted return above risk-free rate.

Return

REPORTED

Cumulative and annualized return.

Max DD

REPORTED

Largest peak-to-trough portfolio decline.

Requested Addition

Sortino Ratio is requested in the topic list but is not currently produced by ``compute_metrics()``.

Latest PSO Results

50.85%

CUM RETURN

PSO Portfolio, with cost, full period.

1.4564

SHARPE RATIO

Mean across 2 trials, std 0.3083.

7.43%

SPY CUM RETURN

Benchmark full-period cumulative return.

Strategy	Cum Return	Ann Return	Sharpe	Max DD
PSO Portfolio	50.85% +- 9.11%	39.65% +- 4.98%	1.4564 +- 0.3083	-21.97% +- 3.05%
LW Div Cluster	45.72% +- 10.05%	36.02% +- 6.50%	1.4161 +- 0.1988	-18.05% +- 0.97%
SPY Benchmark	7.43%	8.10%	0.1949	-19.21%

Latest ACO+EBGWO Results

40.90%

BEST CUM RETURN

ACO Selected, with cost, full period.

1.2425

SHARPE RATIO

Mean across 2 trials, std 0.4215.

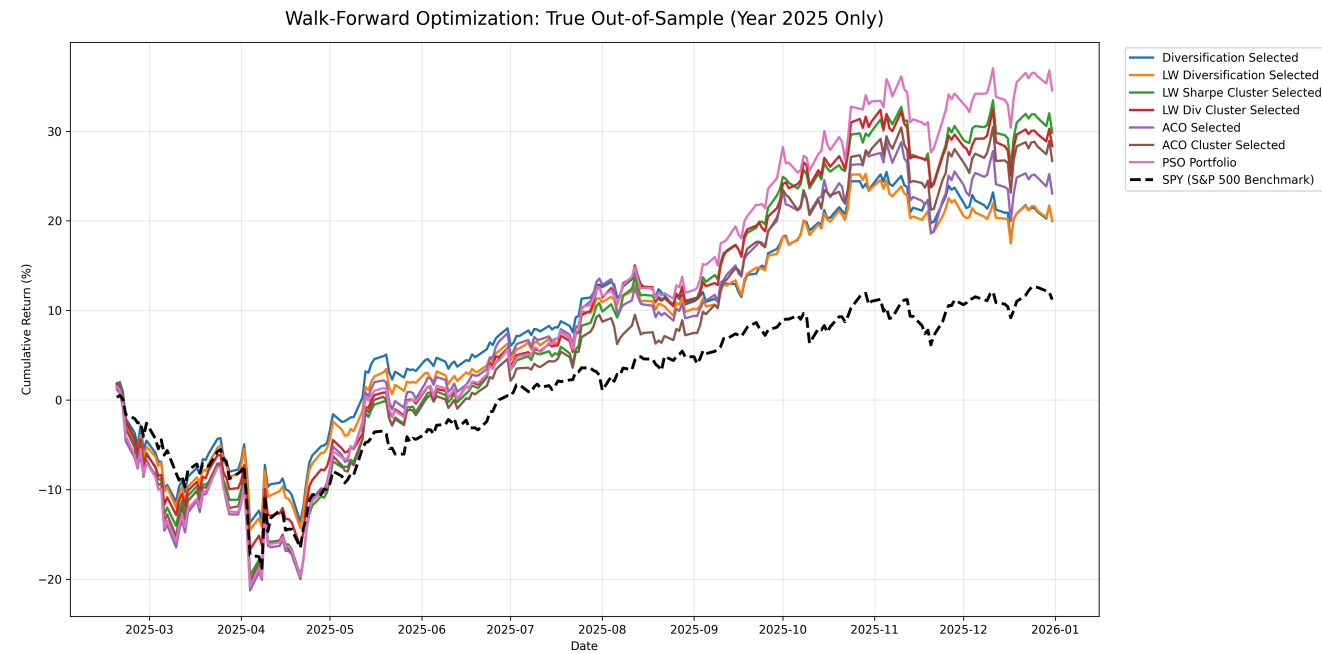
16.43%

DYNAMIC ACO+EBGWO

Higher instability: std 21.98% cumulative return.

Strategy	Cum Return	Ann Return	Sharpe	Max DD
ACO Selected	40.90% +- 11.75%	33.31% +- 6.88%	1.2425 +- 0.4215	-17.75% +- 5.43%
LW Sharpe Cluster	35.51% +- 11.71%	29.30% +- 6.59%	1.2577 +- 0.6371	-16.86% +- 8.94%
SPY Benchmark	7.43%	8.10%	0.1949	-19.21%

2025 Out-of-Sample View



PSO 2025 With Cost

- Cumulative return: 34.53% +- 5.13%.
- Annual return: 36.05% +- 3.73%.
- Sharpe Ratio: 1.2721 +- 0.2522.
- SPY 2025 cumulative return: 11.21%.

Key Findings and Next Steps

Findings

- PSO is the strongest latest executed optimizer.
- ACO Selected leads the ACO+EBGWO run.
- Both outperform SPY on cumulative return and Sharpe.
- Transaction costs reduce results slightly but do not change the PSO leader.

To Align With Rubric

- Add Sortino Ratio to reports.
- Set explicit 4-year train / 1-year test walk-forward.
- Enforce max 15 assets.
- Add Chinese, ETC, and Crypto tickers if those markets are mandatory.
- Add short-selling only if required as a bonus feature.